

1 Riemann-Roch Theorem

Let M be a compact Riemann surface, define

$\text{Div}(M) :=$ free Abelian group generated by points of M with coefficients in $\mathbb{Z}$

an element $D \in \text{Div}(M)$ is called a **divisor** on M , the set $\text{Div}(M)$ forms an Abelian group under the formal addition. An element $D \triangleq \sum_j n_j p_j \in \text{Div}(M)$ is called **effective** if $n_j \geq 0$. And the **degree of** D is defined by

$$\deg D := \sum_j n_j$$

For a meromorphic function $f \in \mathcal{M}(M)$ on M , the **divisor of** f , which is denoted by (f) , is defined as

$$(f) := \sum_{i=1}^k n_i p_i - \sum_{j=1}^l m_j q_j$$

where $\{p_1, \dots, p_k\}$ is the set of zeros of f and $\{q_1, \dots, q_l\}$ is the set of poles of f and the integers n_i, m_j are their orders respectively. We define

$$\text{PDiv}(M) := \{D \in \text{Div}(M) \mid \exists f \in \mathcal{M}(M) \text{ such that } D = (f)\}$$

which is a subgroup of $\text{Div}(M)$ since $(fg) = (f) + (g)$, an element $D \in \text{PDiv}(M)$ is called a **principal divisor** on M . For $D_1, D_2 \in \text{Div}(M)$, we say that D_1 and D_2 are **linearly equivalent** if $D_1 - D_2 \in \text{PDiv}(M)$. We set

$$\text{Cl}(M) := \text{Div}(M)/\text{PDiv}(M)$$

which is called the **divisor class group of** M .

Still let M be a compact Riemann surface, let $L \rightarrow M$ be a holomorphic line bundle on M and let $\{s\}$ be a holomorphic local frame of L on $U \subseteq M$, a **meromorphic section** $\sigma \in \mathcal{M}(M; L)$ can be locally expressed as

$$\sigma|_U(z) \triangleq f_U(z)s(z) \quad (\forall z \in U)$$

with some $f_U \in \mathcal{M}(U)$. We define the **divisor of** σ on U to be $(\sigma|_U) := (f_U)$ and hence we get (σ) . (So, such divisor is effective $\Leftrightarrow f_U \in \mathcal{O}(U)$ for an open cover $\{U\}$ of M , i.e., $\sigma \in \mathcal{O}(M; L)$ is a holomorphic section)

Recall: For a complex manifold M , we have the **Picard group** which is defined by

$$\text{Pic}(M) := \{L \rightarrow M \mid L \text{ is a holomorphic line bundle}\} / \sim$$

where the equivalence relation “ $\sim$ ” is given by $L_1 \sim L_2 \Leftrightarrow L_1 \cong L_2$. The multiplication and inverse on $\text{Pic}(M)$ is given by

$$[L_1] \cdot [L_2] := [L_1 \otimes L_2], \quad [L]^{-1} = [L^*]$$

we may denote $L = [L] \in \text{Pic}(M)$ if there is no confusion. The identity element in $\text{Pic}(M)$ consists of the set of trivial line bundles.

Theorem 1. Let M be a compact Riemann surface. Then, for any $D \in \text{Div}(M)$, $\exists! L_D \in \text{Pic}(M)$ and $\exists \sigma \in \mathcal{M}(M; L_D)$ such that

$$D = (\sigma)$$

furthermore, if $\sigma_1, \sigma_2 \in \mathcal{M}(M; L_D)$ such that $D = (\sigma_1) = (\sigma_2)$, then $\exists k \in \mathbb{C}$ a constant such that $\sigma_1 = k\sigma_2$.

Proof: Write $D \triangleq \sum_{\alpha=1}^k n_\alpha p_\alpha$, for each $n_\alpha \neq 0$, we can choose a neighborhood U_α of p_α and choose $f_{U_\alpha} \in \mathcal{M}(U_\alpha)$ such that $f_{U_\alpha} \in \mathcal{O}^*(U_\alpha \setminus \{p_\alpha\})$ and p_α would be its zero of order n_α if $n_\alpha > 0$ and would be its pole of order $-n_\alpha$ if $n_\alpha < 0$. Denote $U_0 := M \setminus \{p_1, \dots, p_k\}$ and define $f_{U_0} := 1$ on U_0 and hence we have an open cover $\{U_0, U_{\alpha_1}, \dots, U_{\alpha_k}\}$ of M . Whenever $U_\alpha \cap U_\beta \neq \emptyset$ for some $\alpha, \beta \in \{1, 2, \dots, k\}$, we set $\tau_{\alpha\beta} := f_{U_\alpha}/f_{U_\beta}$, thus we get

$$\tau_{\alpha\beta} : U_\alpha \cap U_\beta \longrightarrow GL(1; \mathbb{C}) = \mathbb{C}^*$$

$$p \mapsto \tau_{\alpha\beta}(p) = \frac{f_{U_\alpha}(p)}{f_{U_\beta}(p)}$$

which satisfies

$$\tau_{\alpha\gamma} = \tau_{\alpha\beta}\tau_{\beta\gamma} \quad \text{whenever } U_\alpha \cap U_\beta \cap U_\gamma \neq \emptyset$$

Thus by the theory of vector bundles (see [2]), $\exists L_D \in \text{Pic}(M)$ with local holomorphic frame $\{s_\alpha\}$ on U_α with transition function $\tau_{\alpha\beta}$. Then we set

$$\sigma|_{U_\alpha} := f_{U_\alpha} s_\alpha$$

hence by the definition we immediately get $\sigma \in \mathcal{M}(M; L_D)$ and $(\sigma) = D$.

If $\sigma_1, \sigma_2 \in \mathcal{M}(M; L_D)$ both satisfies $D = (\sigma_1) = (\sigma_2)$, then by the construction above, we see that σ_1/σ_2 has only removable singularities, i.e., σ_1/σ_2 can be extended holomorphically to the entire of M , since M is compact, by the maximum principle of modules, there is a constant $k \in \mathbb{C}$ such that $\sigma_1/\sigma_2 \equiv k$.

Suppose $L' \in \text{Pic}(M)$ also satisfies the condition, i.e., $\exists \sigma' \in \mathcal{M}(M; L')$ such that $D = (\sigma')$. We show that $L_D \cong L'$: Choose an open cover $\{U_\alpha\}$ which trivialize both L_D and L' and let $\{s_\alpha\}, \{s'_\alpha\}$ be the local holomorphic frame for L_D, L' on U_α respectively, and their transition functions are denoted as $\tau_{\alpha\beta}, \tau'_{\alpha\beta}$ respectively, i.e.,

$$s_\beta = \tau_{\alpha\beta} s_\alpha, \quad s'_\beta = \tau'_{\alpha\beta} s'_\alpha$$

then we have

$$\sigma|_{U_\alpha} \triangleq f_{U_\alpha} s_\alpha, \quad \sigma'|_{U_\alpha} \triangleq f'_{U_\alpha} s'_\alpha$$

for some $f_{U_\alpha}, f'_{U_\alpha} \in \mathcal{M}(U_\alpha)$, so by our construction, we see that

$$f_{U_\alpha} s_\alpha = f_{U_\beta} s_\beta = f_{U_\beta} \tau_{\alpha\beta} s_\alpha \quad (U_\alpha \cap U_\beta \neq \emptyset)$$

i.e.,

$$f_{U_\alpha} = f_{U_\beta} \tau_{\alpha\beta} \quad (U_\alpha \cap U_\beta \neq \emptyset)$$

similarly, we also have

$$f'_{U_\alpha} = f'_{U_\beta} \tau'_{\alpha\beta} \quad (U_\alpha \cap U_\beta \neq \emptyset)$$

Thus, on $U_\alpha \cap U_\beta$, we have

$$\tau_{\alpha\beta} = \frac{f'_{U_\alpha} f_{U_\beta} \tau_{\alpha\beta}}{f'_{U_\alpha} f_{U_\beta}} = \frac{f'_{U_\alpha} f_{U_\alpha}}{f'_{U_\alpha} f_{U_\beta}} = \frac{f'_{U_\beta} \tau'_{\alpha\beta} f_{U_\alpha}}{f'_{U_\alpha} f_{U_\beta}} = \frac{f'_{U_\beta}}{f_{U_\beta}} \tau'_{\alpha\beta} \frac{f_{U_\alpha}}{f'_{U_\alpha}}$$

note that since $(\sigma') = (\sigma)$, so $(\sigma'|_{U_\alpha}) = (\sigma|_{U_\alpha})$, which implies that $(f|_{U_\alpha}) = (f'|_{U_\alpha})$, hence $\exists \lambda_\alpha \in \mathcal{O}^*(U_\alpha \cap U_\beta)$ such that $\lambda_\alpha = f_{U_\alpha}/f'_{U_\alpha}$. So back to our computation, we get

$$\tau_{\alpha\beta} = \lambda_\beta^{-1} \tau'_{\alpha\beta} \lambda_\alpha$$

so by the isomorphism criterion of vector bundles (see [2]), we know that $L' \cong L_D$, i.e. $[L'] = [L_D] \in \text{Pic}(M)$. ■

Remark: The holomorphic line bundle $L_D \in \text{Pic}(M)$ in this theorem is called **associated line bundle for D** .

Proposition 1. For a compact Riemann surface M , the map

$$\Phi : \text{Div}(M) \longrightarrow \text{Pic}(M)$$

$$D \mapsto [L_D]$$

is a group homomorphism with $\ker \Phi = \text{PDiv}(M)$. i.e., we have

$$L_{D_1} \otimes L_{D_2} \cong L_{D_1+D_2}, \quad L_D^* \cong L_{-D}$$

Proof: Take $\forall D_1, D_2 \in \text{Div}(M)$, by the theorem above, $\exists L_{D_1}, L_{D_2} \in \text{Pic}(M)$ and $\exists \sigma_1 \in \mathcal{M}(M; L_{D_1}), \sigma_2 \in \mathcal{M}(M; L_{D_2})$ such that $D_1 = (\sigma_1), D_2 = (\sigma_2)$ note that

$$\sigma_1 \otimes \sigma_2 \in \mathcal{M}(M; L_{D_1} \otimes L_{D_2})$$

and

$$D_1 + D_2 = (\sigma_1 \otimes \sigma_2)$$

hence the uniqueness in the theorem above ensures that

$$L_{D_1} \otimes L_{D_2} \cong L_{D_1+D_2}$$

i.e., $\Phi(D_1 + D_2) = [L_{D_1+D_2}] = [L_{D_1} \otimes L_{D_2}] = [L_{D_1}] \cdot [L_{D_2}] = \Phi(D_1)\Phi(D_2)$, so Φ is a group homomorphism;

Suppose $D \in \ker \Phi$, hence $[L_D] = 0 \in \text{Pic}(M)$, i.e., $L_D \cong M \times \mathbb{C}$. And $\exists \sigma \in \mathcal{M}(M; L_D)$ such that $D = (\sigma)$. Since $\sigma \in \mathcal{M}(M; M \times \mathbb{C}) = \mathcal{M}(M)$, i.e., σ can be viewed as a meromorphic function on M , hence $D \in \text{PDiv}(M)$; Conversely, if $\exists f \in \mathcal{M}(M)$ such that $D = (f)$, note that $f \in \mathcal{M}(M) = \mathcal{M}(M; M \times \mathbb{C})$, so by the uniqueness in the theorem above again, we have $L_D \cong M \times \mathbb{C}$. Thus $\text{PDiv}(M) = \ker \Phi$. ■

Corollary 1. For a compact Riemann surface M , the map

$$\Phi : \text{Div}(M) \longrightarrow \text{Pic}(M)$$

$$D \mapsto [L_D]$$

descends to an injective homomorphism

$$\tilde{\Phi} : \text{Cl}(M) \hookrightarrow \text{Pic}(M)$$

$$[D] \mapsto [L_D]$$

Proof: By the proposition above and the definition that $\text{Cl}(M) := \text{Div}(M)/\text{PDiv}(M)$. ■

We will show in the sequel that the map

$$\tilde{\Phi} : \text{Cl}(M) \longrightarrow \text{Pic}(M)$$

is also surjective and thus this is actually an isomorphism. For a compact Riemann surface M , the bundle $L_{\{p\}}$ associated to a single point $p \in M$ is called a **point bundle**. Note that the hypersurface on M is just a finite collection of finite points on M .

Recall: Let $\pi : E \longrightarrow M$ be a holomorphic vector bundle, we denote $\mathcal{O}(E)$ to be the sheaf on M such that for any open subset $U \subseteq M$

$$\mathcal{O}(E)(U) := \mathcal{O}(U; E)$$

the local holomorphic sections of E on U .

Recall: For a sheaf $\mathcal{A}$ on the complex manifold M , if $\dim H^k(M; \mathcal{A})$ are all finite, then the **Euler characteristic of $\mathcal{A}$** is given by $\chi(\mathcal{A}) := \sum_j (-1)^j \dim H^j(M; \mathcal{A})$.

Recall: For a manifold M , we have

$$H_{\text{dR}}^k(M; \mathbb{C}) \cong H_{\text{Sing}}^k(M; \mathbb{C}) \cong H^k(M; \underline{\mathbb{C}})$$

which can be seen in [2], where $\underline{\mathbb{C}}$ is constant sheaf, i.e.,

$$\underline{\mathbb{C}}(U) := \{f : U \longrightarrow \mathbb{C} \mid f \text{ is locally constant}\}$$

Recall: Recall that for a compact manifold M , $\dim H_{\text{dR}}^k(M; \mathbb{C}) < \infty$. (see [2])

Recall: For a subset $S \subseteq M$, we denote the sheaf $\mathcal{I}_S$ on M to be

$$\mathcal{I}_S(U) := \{f \in \mathcal{O}(U) \mid f|_{S \cap U} \equiv 0\}$$

$\mathcal{I}_S$ is called the **ideal sheaf of S** . More generally, for a holomorphic vector bundle $E \longrightarrow M$, define the sheaf $\mathcal{I}_S(E)$ to be

$$\mathcal{I}_S(E)(U) := \{\sigma \in \mathcal{O}(U; E) \mid \sigma|_{S \cap U} \equiv 0\}$$

Recall: For an Abelian group G and a point $p \in M$, define the sheaf G_p on M to be

$$G_p(U) := \begin{cases} G, & p \in U \\ \{0\}, & p \notin U \end{cases}$$

the restriction map ρ_{UV} is given by

$$\rho_{UV} := \begin{cases} \text{Id}_G, & p \in V \subseteq U \\ 0, & \text{otherwise} \end{cases}$$

such sheaf is called a **skyscraper sheaf**.

Lemma 1. For a compact Riemann surface M and $L \in \text{Pic}(M)$. Then for $\forall p \in M$, we have

$$\chi(\mathcal{O}(L \otimes L_{\{p\}})) = \chi(\mathcal{O}(L)) + 1$$

$$\chi(\mathcal{O}(L \otimes L_{\{p\}}^*)) = \chi(\mathcal{O}(L)) - 1$$

Sketch of Proof: Consier the exact sequence of sheaves on M

$$0 \longrightarrow \mathcal{I}_{\{p\}}(L) \xrightarrow{\iota} \mathcal{O}(L) \xrightarrow{e} (L_p)_p \longrightarrow 0$$

where $(L_p)_p$ is the skyscraper sheaf whose stalk at p is the fiber L_p with all other stalks zero. The sheaf morphism e is defined by

$$e_U(\sigma) := \sigma(p) \text{ (if } p \in U), \quad e_U = 0 \text{ (otherwise)}$$

it can be checked that $\mathcal{I}_{\{p\}}(L) \cong \mathcal{O}(L \otimes L_{\{p\}}^*)$ and also we know that skyscraper sheaf $(L_p)_p$ is a fine sheaf, which implies that $(L_p)_p$ is a fine sheaf, i.e., $H^k(M; (L_p)_p) = 0$ for $\forall k > 0$. Note that $H^0(M; (L_p)_p) \cong (L_p)_p(M) = L_p$, so we get

$$\chi((L_p)_p) = \dim L_p = 1.$$

By the exactness of the sequence of sheaves, we have

$$\chi(\mathcal{O}(L)) = \chi(\mathcal{I}_{\{p\}}(L)) + \chi((L_p)_p)$$

since $\mathcal{I}_{\{p\}}(L) \cong \mathcal{O}(L \otimes L_{\{p\}}^*)$, we have $\chi(\mathcal{I}_{\{p\}}(L)) = \chi(\mathcal{O}(L \otimes L_{\{p\}}^*))$, thus

$$\chi(\mathcal{O}(L)) = \chi(\mathcal{O}(L \otimes L_{\{p\}}^*)) + 1$$

i.e.

$$\chi(\mathcal{O}(L \otimes L_{\{p\}}^*)) = \chi(\mathcal{O}(L)) - 1$$

which is the second equation. Replace L by $L \otimes L_{\{p\}}$, we get

$$\chi(\mathcal{O}(L \otimes L_{\{p\}})) = \chi(\mathcal{O}(L \otimes L_{\{p\}} \otimes L_{\{p\}}^*)) + 1 = \chi(\mathcal{O}(L)) + 1$$

which is the first equation. ■

Corollary 2. For a compact Riemann surface M and $L \in \text{Pic}(M)$. Then for $\forall p \in M$ and $\forall k \in \mathbb{N}$, we have

$$\begin{aligned}\chi(\mathcal{O}(L \otimes L_{\{p\}}^k)) &= \chi(\mathcal{O}(L)) + k \\ \chi(\mathcal{O}(L \otimes (L_{\{p\}}^*)^k)) &= \chi(\mathcal{O}(L)) - k\end{aligned}$$

Proof: By induction. ■

Recall: For a complex manifold M , the sheaf Ω^p on M is defined by

$$\Omega^p(U) := \mathcal{O}(\Lambda^{p,0}(T^*U)_{\mathbb{C}}) = \mathcal{O}(\wedge^p(T^{(1,0)*}U))$$

and hence $\Omega^0(U) = \mathcal{O}(U)$. The **Dolbeault theorem** says that

$$H^q(M; \Omega^p) \cong H^{p,q}(M)$$

and hence particularly

$$H^q(M; \mathcal{O}) \cong H^{0,q}(M)$$

Generally, for a holomorphic vector bundle $E \rightarrow M$, the sheaf $\Omega^p(E)$ is given by

$$\Omega^p(E)(U) := \mathcal{O}(\Lambda^{p,0}(T^*U)_{\mathbb{C}} \otimes E) = \mathcal{O}(\wedge^p(T^{(1,0)*}U) \otimes E)$$

hence $\Omega^0(E) = \mathcal{O}(E)$ and the Dolbeault theorem generalize to

$$H^q(M; \Omega^p(E)) \cong H^{p,q}(M; E)$$

$$H^q(M; \mathcal{O}(E)) \cong H^{0,q}(M; E)$$

Theorem 2. For a compact Riemann surface M , the map

$$\tilde{\Phi} : \text{Cl}(M) \rightarrow \text{Pic}(M)$$

$$[D] \mapsto [L_D]$$

is an isomorphism.

Proof: Since

$$H^q(M; \mathcal{O}(L)) \cong H^{0,q}(M; L)$$

thus $H^q(M; \mathcal{O}(L)) = 0$ for $\forall q \geq 2$ and therefore we have

$$\chi(\mathcal{O}(L)) = \dim H^0(M; \mathcal{O}(L)) - \dim H^1(M; \mathcal{O}(L))$$

for any point $p \in M$ and any $k \in \mathbb{N}$, by the corollary above, we get

$$\dim H^0(M; L \otimes L_{\{p\}}^k) = \dim \mathcal{O}(M; L \otimes L_{\{p\}}^k) = \chi(\mathcal{O}(L \otimes L_{\{p\}}^k)) + \dim H^1(M; L \otimes L_{\{p\}}^k)$$

$$\geq \chi(\mathcal{O}(L \otimes L_{\{p\}}^k)) = \chi(\mathcal{O}(L)) + k$$

hence for $k \in \mathbb{N}$ large enough, we have $\dim \mathcal{O}(M; L \otimes L_{\{p\}}^k) > 0$, which means $\exists$ non-trivial $\sigma \in \mathcal{O}(M; L \otimes L_{\{p\}}^k)$, if we denote $D_1 := (\sigma)$, then by theorem 1, we get

$$L_{D_1} \cong L \otimes L_{\{p\}}^k$$

If we take $D_2 := kp \in \text{Div}(M)$, then we have

$$L_{\{p\}}^k \cong L_{D_2}$$

and hence

$$(L_{\{p\}}^k)^* \cong L_{-D_2}$$

so we have

$$L \cong L \otimes (L_{\{p\}}^k \otimes (L_{\{p\}}^k)^*) \cong (L \otimes L_{\{p\}}^k) \otimes (L_{\{p\}}^k)^* \cong L_{D_1} \otimes L_{-D_2} \cong L_{D_1 - D_2}$$

so this means for any $[L] \in \text{Pic}(M)$, it has a preimage $[D_1 - D_2] \in \text{Cl}(M)$ under $\tilde{\Phi}$. Combine Corollary 1, so $\tilde{\Phi}$ is an isomorphism. ■

Recall: (**Serre duality**) For a compact complex manifold M of dimension n , let $E \rightarrow M$ be a holomorphic vector bundle on M , then we have a non-degenerate bilinear form

$$\int_M : H^{p,q}(M; E) \times H^{n-p, n-q}(M; E^*) \rightarrow \mathbb{C}$$

$$([\alpha], [\beta]) \mapsto \int_M \alpha \wedge \beta$$

hence it induces the following isomorphisms

$$H^{p,q}(M; E)^* \cong H^{n-p, n-q}(M; E^*)$$

by Dolbeault theorem $H^{p,q}(M; E) \cong H^q(M; \Omega^p(E))$, so we have

$$H^q(M; \Omega^p(E))^* \cong H^{n-q}(M; \Omega^{n-p}(E^*))$$

and particularly if we take $p = 0$, we get

$$H^q(M; \mathcal{O}(E))^* \cong H^{n-q}(M; \mathcal{O}(K_M \otimes E^*))$$

where $K_M := \Lambda^{n,0}((T^*M)_{\mathbb{C}})$ is the canonical bundle. Thus we have

$$h^{p,q}(M) = h^{n-p, n-q}(M)$$

Recall: For a connected compact Riemann surface M , since M is of course Kahler, we know that there is the Hodge decomposition

$$H_{\text{dR}}^1(M; \mathbb{C}) \cong H^{1,0}(M) \oplus H^{0,1}(M)$$

and by Serre duality we know that $h^{1,0}(M) = h^{0,1}(M)$, hence $b_1(M) = h^{1,0}(M) + h^{0,1}(M) = 2h^{1,0}(M)$ is even. The **genus** of M is thus defined by $g(M) := \frac{1}{2} \dim H_{\text{dR}}^1(M) = \frac{1}{2} b_1(M) = h^{1,0}(M)$.

Recall: For a smooth manifold M , there is the exact sequence of sheaves on M

$$0 \longrightarrow \underline{\mathbb{Z}} \xrightarrow{\iota} \mathcal{E} \xrightarrow{\varepsilon} \mathcal{E}^* \longrightarrow 0$$

where $\mathcal{E}(U) := \{f : U \rightarrow \mathbb{C} \mid f \in C^\infty(U; \mathbb{C})\}$, $\mathcal{E}^*(U) := \{f \in \mathcal{E}(U) \mid f \text{ non-vanish on } U\}$ and the sheaf morphism $\varepsilon_U(f) := \exp(2\pi\sqrt{-1}f)$. Hence it induces the long exact sequences of cohomology

$$\dots \longrightarrow H^1(M; \mathcal{E}) \longrightarrow H^1(M; \mathcal{E}^*) \xrightarrow{\delta_*} H^2(M; \underline{\mathbb{Z}}) \longrightarrow H^2(M; \mathcal{E}) \longrightarrow \dots$$

In the sheaf theory, we know that

$$H^1(M; \mathcal{E}^*) \cong \text{Pic}(M)$$

then we define

$$c(L) := -\delta_*([L]) \in H^2(M; \underline{\mathbb{Z}}) \quad (\forall L \in \text{Pic}(M))$$

Particularly, if M is a compact Riemann surface, then for $L \in \text{Pic}(M)$, we define the **degree of L** to be

$$\deg L := \langle c(L), [M] \rangle$$

where $[M] \in H_2(M)$ is the fundamental class of M and $\langle \cdot, \cdot \rangle$ is the Kronecker pairing, note that $c(L) \in H^2(M; \underline{\mathbb{Z}}) \cong H_{\text{Sing}}^2(M; \mathbb{Z})$.

The important result about the degree of $L \in \text{Pic}(M)$ on a compact connected Riemann surface M is the following

- (1) $L_1 \cong L_2 \Leftrightarrow \deg L_1 = \deg L_2$
- (2) $\deg : \text{Pic}(M) \rightarrow \mathbb{Z}$ is surjective
- (3) $\deg L_D = \deg D$ for $\forall D \in \text{Div}(M)$.

Theorem 3. (Riemann-Roch) Let M be a connected compact Riemann surface with genus $g := g(M)$. Then for any $L \in \text{Pic}(M)$, we have

$$\dim \mathcal{O}(M; L) = \deg L + 1 - g + \dim \mathcal{O}(M; K \otimes L^*)$$

where K is the canonical line bundle, i.e., $K = \Lambda^{1,0}((T^*M)_{\mathbb{C}})$ in this case. It is also equivalent to

$$\chi(\mathcal{O}(L)) = \deg L + 1 - g$$

Proof: In the proof of Theorem 2, we've seen that

$$\chi(\mathcal{O}(L)) = \dim H^0(M; \mathcal{O}(L)) - \dim H^1(M; \mathcal{O}(L))$$

by Serre duality, we have

$$\dim H^1(M; \mathcal{O}(L)) = \dim H^0(M; \mathcal{O}(K \otimes L^*))$$

and note that $H^0(M; \mathcal{O}(L)) \cong \mathcal{O}(L)(M) = \mathcal{O}(M; L)$, so we get

$$\chi(\mathcal{O}(L)) = \dim \mathcal{O}(M; L) - \dim \mathcal{O}(M; K \otimes L^*)$$

So the equaion $\dim \mathcal{O}(M; L) = \deg L + 1 - g + \dim \mathcal{O}(M; K \otimes L^*)$ is equivalent to $\chi(\mathcal{O}(L)) = \deg L + 1 - g$. We will prove the latter:

Note that $K = \Lambda^{1,0}((T^*M)_{\mathbb{C}})$, hence $\mathcal{O}(M; K) = \mathcal{O}(M; \Lambda^{1,0}((T^*M)_{\mathbb{C}}))$, i.e., the space of holomorphic $(1,0)$ -forms on M . For any $\alpha \in \mathcal{O}(M; K)$, it's obvious that $\bar{\partial}\alpha = 0$ and $\bar{\partial}^*\alpha = 0$, so this means that

$$\alpha \in \mathcal{H}^{1,0}(M) := \ker \left(\Delta_{\bar{\partial}} : \Omega^{1,0}(M) \longrightarrow \Omega^{1,0}(M) \right)$$

hence $\mathcal{O}(M; K) \subseteq \mathcal{H}^{1,0}(M)$ and it's clear that $\mathcal{H}^{1,0}(M) \subseteq \mathcal{O}(M; K)$, therefore by the Hodge theorem on compact Kahler manifolds, we get

$$\dim \mathcal{O}(M; K) = \dim H^{1,0}(M) = \dim H^{1,0}(M) = h^{1,0}(M) = g$$

Now, by Theorem 2, we know that $\exists D \in \text{Div}(M)$ such that $L \cong L_D$ and $\deg L = \deg D$.

First, we assume that D is effective, i.e., $D = \sum_j n_j p_j$ with $n_j \geq 0$ and hence $\deg D = \sum_j n_j$.

We prove by induction on $\deg D$:

1° if $\deg D = 0$, then this means that $D = 0$, it's clear that L_D is a trivial bundle hence so is L , in which case we would get $\dim \mathcal{O}(M; L) = 1$ and $\dim \mathcal{O}(M; K \otimes L^*) = \dim \mathcal{O}(M; K) = g$, the conclusion holds.

2° assume the conclusion holds for any $\deg D \leq k - 1$.

3° if $\deg D = k$, since D is effective, we may express D as

$$D \triangleq D' + p_0$$

with $\deg D' = k - 1$ and $p_0 \in M$. Then by Porposition 1, we get $L_D \cong L_{D'} \otimes L_{\{p_0\}}$ and also we get $\deg(L_D) = \deg D = \deg D' + 1 = \deg L_{D'} + 1$. By induction hypothesis 2°, we have

$$\chi(\mathcal{O}(L_{D'})) = \deg L_{D'} + 1 - g$$

then by Lemma 1, we get

$$\chi(\mathcal{O}(L_D)) = \chi(\mathcal{O}(L_{D'} \otimes L_{\{p_0\}})) = \chi(\mathcal{O}(L_{D'})) + 1 = (\deg L_{D'} + 1) + 1 - g = \deg L_D + 1 - g$$

which completes the proof in the case that D is effective.

Now, suppose D is in the general case, then we can write D as

$$D \triangleq D_1 - D_2$$

with D_1, D_2 effective, hence we get

$$L_D \cong L_{D_1} \otimes L_{-D_2} \cong L_{D_1} \otimes L_{D_2}^*$$

so by Lemma 1 agian, we can get

$$\chi(\mathcal{O}(L_D)) = \chi(\mathcal{O}(L_{D_1} \otimes L_{D_2}^*)) = \chi(\mathcal{O}(L_{D_1})) - \deg L_{D_2}$$

Thus, by the discussion in the effective case above, we have

$$\chi(\mathcal{O}(L_D)) = \deg L_{D_1} + 1 - g - \deg L_{D_2} = (\deg L_{D_1} - \deg L_{D_2}) + 1 - g$$

note that $\deg L_{D_1} - \deg L_{D_2} = \deg D_1 - \deg D_2 = \deg D = \deg L_D$, so we finnaly get

$$\chi(\mathcal{O}(L)) = \chi(\mathcal{O}(L_D)) = \deg L_D + 1 - g = \deg L + 1 - g$$

which completes the proof of the theorem. ■

2 Riemann-Roch-Hirzebruch Theorem

There is a generalization of Riemann-Roch theorem to the complex manifolds of higher dimension, which was proved by Hirzebruch.

Consider the Todd function

$$\text{Td}(x) := \frac{1}{1 - \exp(-x)} = 1 + \frac{x}{2} + \sum_{k=1}^{\infty} (-1)^{k+1} \frac{B_{2k}}{(2k)!} x^{2k} = 1 + \frac{x}{2} + \frac{x^2}{12} - \frac{x^4}{120} + \dots$$

where B_k is the Bernoulli numbers. Let $E \rightarrow M$ be a smooth complex vector bundle of rank k on the complex manifold M , given any connection ∇^E on E , then for any smooth local frame $\{e_i\}_{i=1}^k$, we have the connection forms $\omega_i^j \in \Omega_{\mathbb{C}}^1(M)$ and curvature forms $\Omega_i^j \in \Omega_{\mathbb{C}}^2(M)$, i.e.,

$$\nabla^E s_i = \sum_{j=1}^k \omega_i^j \otimes s_j, \quad R^E s_i = \sum_{j=1}^k \Omega_i^j \otimes s_j$$

and denote the curvature matrix $\Omega := (\Omega_i^j)$. If we denote

$$\det \left(\text{Td} \left(\frac{\sqrt{-1}}{2\pi} t \Omega \right) \right) = \det \left(\frac{\frac{\sqrt{-1}}{2\pi} t \Omega}{I_k - \exp(-\frac{\sqrt{-1}}{2\pi} t \Omega)} \right) \triangleq 1 + t \text{Td}_1(\Omega) + t^2 \text{Td}_2(\Omega) + \dots + t^k \text{Td}_k(\Omega)$$

then we get $\text{Td}_j \in \text{Inv}^j(GL_k(\mathbb{C}))$ and $\text{Td}_j(\Omega) \in \Omega_{\mathbb{C}}^{2j}(M)$. The important fact is that by Chern-Weil theory, $\text{Td}_j(\Omega) \in \ker(d|_{\Omega_{\mathbb{C}}^{2j}(M)})$ and the cohomology class $[\text{Td}_j(\Omega)] \in H_{\text{dR}}^{2j}(M; \mathbb{C})$ is independent of the choice of the local frame $\{e_i\}$ and more importantly independent of the choice of the connection ∇^E . Thus we denote

$$\text{Td}_j(E, \nabla^E) := \text{Td}_j(\Omega) \in \Omega_{\mathbb{C}}^{2j}(M), \quad \text{Td}_j(E) := [\text{Td}_j(E, \nabla^E)] \in H_{\text{dR}}^{2j}(M; \mathbb{C})$$

$$\text{Td}(E, \nabla^E) := \sum_{j=0}^k \text{Td}_j(E, \nabla^E) \in \Omega_{\mathbb{C}}^{\text{even}}(M)$$

and define the **total Todd class** of E to be

$$\text{Td}(E) := \sum_{j=0}^k \text{Td}_j(E) \in H_{\text{dR}}^{\text{even}}(M; \mathbb{C})$$

We also denote

$$\text{tr} \left(\exp \left(\frac{t\sqrt{-1}}{2\pi} \Omega \right) \right) \triangleq \sum_{j=0}^{\infty} t^j \text{ch}_j(\Omega)$$

i.e., $\text{ch}_j(\Omega) = \frac{1}{j!} \text{tr} \left\{ \left(\frac{\sqrt{-1}}{2\pi} \Omega \right)^j \right\} \in \Omega_{\mathbb{C}}^{2j}(M)$, which is called the **j-th Chern character** of E , and

$$\text{ch}(E) := \sum_{j=0}^{\infty} \text{ch}_j(E) = \sum_{j=0}^{\infty} \frac{1}{j!} \text{tr} \left\{ \left(\frac{\sqrt{-1}}{2\pi} \Theta \right)^j \right\}$$

is called the **total Chern character** of E .

Theorem 4. (Riemann-Roch-Hirzebruch) Let M be a compact complex manifold and $E \rightarrow M$ is a holomorphic vector bundle on M , then we have

$$\chi(M; E) = \int_M \text{ch}(E) \text{Td}(M)$$

where $\text{Td}(M) := \text{Td}(T^{(1,0)}M, \nabla^{T^{(1,0)}M})$ and $\chi(M; E) = \sum_j (-1)^j \dim H^j(M; E)$ is the Euler characteristic of E .

3 References

- [1] *Introduction to Complex Manifolds.* John M. Lee
- [2] *Differential Forms in Algebraic Topology.* Raoul Bott, Loring W. Tu.
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